

The Playlist Operator:

a matrix that plays music, whose picture is an image of your choice

1 What we want to be true

Take N songs, each a vector of audio samples $x_1, \dots, x_N \in \mathbb{R}^D$ (here $N = 6$, $D \approx 5.5 \times 10^5$ at 1 kHz). We want a single matrix $A \in \mathbb{R}^{D \times D}$ that *is* the playlist:

$$Ax_n = x_{n+1} \quad (n = 1, \dots, N-1), \quad Ax_N = x_1. \quad (1)$$

Feed it any song of the playlist and one matrix–vector product returns the next song, forever, cyclically. Stacking the songs as columns of $X = [x_1 \cdots x_N] \in \mathbb{R}^{D \times N}$ and writing $X_{\text{next}} = [x_2, \dots, x_N, x_1]$ (columns advanced one step), all N conditions in (1) become one matrix equation:

$$AX = X_{\text{next}}. \quad (2)$$

The key count. Equation (2) imposes DN scalar constraints on the D^2 entries of A . Since $N \ll D$, the solution set is a huge affine family: roughly $D^2 - DN \approx D(D - N)$ dimensions of freedom remain. *The songs are low rank* — they span only an N -dimensional subspace $\mathcal{S} = \text{span}(x_1, \dots, x_N) \subset \mathbb{R}^D$ — so (2) pins down how A acts on $\mathcal{S}$ and says *nothing* about how it acts on the orthogonal complement $\mathcal{S}^\perp$. That freedom is where the artwork lives.

2 Solving $AX = X_{\text{next}}$, and why $G = X^\top X$ appears

Coefficients in the song basis. Given any vector $v \in \mathbb{R}^D$, its best approximation inside $\mathcal{S}$ is a combination Xc with coefficients $c \in \mathbb{R}^N$ chosen by least squares:

$$\min_c \|Xc - v\|^2 \implies X^\top Xc = X^\top v \implies c = G^{-1}X^\top v, \quad G \equiv X^\top X \in \mathbb{R}^{N \times N}. \quad (3)$$

G is the *Gram matrix* of the songs: $G_{ij} = x_i^\top x_j$ collects all pairwise song correlations. It is invertible exactly when the songs are linearly independent (for real recordings they always are; our measured condition number is 1.01, i.e. the songs are nearly orthogonal). The operator

$$P = XG^{-1}X^\top \quad (\text{size } D \times D, \text{ rank } N) \quad (4)$$

is the *orthogonal projector onto $\mathcal{S}$* : it reads a vector, computes its song coefficients (3), and reassembles the part of the vector that lies in song space. Its complement

$$P_\perp = I - XG^{-1}X^\top \quad (5)$$

keeps only what is orthogonal to every song; in particular $P_\perp x_n = 0$ for all n .

The minimum-norm solution. Now solve (2). Try $A_0 = X_{\text{next}} G^{-1} X^\top$ and check it:

$$A_0 X = X_{\text{next}} \underbrace{G^{-1} X^\top X}_{= G^{-1} G = I_N} = X_{\text{next}}. \quad \checkmark \quad (6)$$

This is why G^{-1} is the essential quantity: it *unmixes* the song correlations so that each x_n is mapped to exactly its own successor and nothing else. Read $A_0 = X_{\text{next}}(G^{-1}X^\top)$ right to left: correlate the input against all songs ($X^\top$), unmix (G^{-1}), emit the corresponding successors (X_{next}). A_0 is the *smallest* solution (minimum Frobenius norm), and it annihilates everything orthogonal to the songs.

The full solution family. If A and A' both satisfy (2), their difference satisfies $(A - A')X = 0$, i.e. it kills song space — which means it can be written as (anything) $\times P_\perp$. Hence *every* exact playlist operator has the form

$$A = A_0 + Z P_\perp, \quad Z \in \mathbb{R}^{D \times D} \text{ completely free.} \quad (7)$$

The free part is invisible to playback: $Z P_\perp x_n = Z 0 = 0$. Playback exactness is enforced *by construction*, not by optimization; our code verifies $\max_n \|Ax_n - x_{n+1}\| / \|x_{n+1}\| \approx 10^{-14}$ (floating-point roundoff) for every candidate artwork.

3 Rendering: how a $D \times D$ matrix becomes a $P \times P$ picture

A has $D^2 \approx 3 \times 10^{11}$ entries — never materialized. It is kept in factored form $A = LR^\top$ with $L = [X_{\text{next}} G^{-1} \mid U]$, $R = [X \mid P_\perp V]$ when $Z = UV^\top$ is low rank. A $P \times P$ picture must summarize $k \times k$ blocks of matrix entries, $k = \lfloor D/P \rfloor$. Define the block-pooling operator $\Pi \in \mathbb{R}^{P \times D}$ (row p averages samples in block p). Two natural per-block statistics:

$$\text{block mean: } M = \Pi A \Pi^\top = (\Pi L)(\Pi R)^\top \quad (\text{linear in } A) \quad (8)$$

$$\text{block energy: } E_{pq} = \frac{1}{k^2} \sum_{i \in p, j \in q} A_{ij}^2 = \text{tr}(G_p H_q) \quad (\text{quadratic in } A) \quad (9)$$

with small per-block Gram matrices $G_p = \frac{1}{k} \sum_{i \in p} L_i L_i^\top$, $H_q = \frac{1}{k} \sum_{j \in q} R_j R_j^\top$ ($L_i = i$ -th row of L). Both stream through the factors; cost scales with pixels, never D^2 .

Why two channels? The matrix is one object; these are two different summaries of the same A , and the choice is purely presentational. They show different things: audio oscillates around zero, so inside any block the entries of A_0 cancel and its *mean* picture is nearly blank — but their *squares* do not cancel, so the energy picture reveals the song-block structure (loudness correlations, silence bands). Conversely, the smooth free part Z shows up strongly in the mean channel. For the embedded artwork below we display the mean channel: it is the channel that is linear in A , hence fully controllable.

4 Making the picture be an image of your choice

Here is the reframing. Because (8) is *linear* in A and the free part of A is enormous, the picture is almost unconstrained. Put $Z = UV^\top$ (rank q) into (8):

$$M = \underbrace{\Pi A_0 \Pi^\top}_{M_0 \text{ (fixed by the music)}} + (\Pi U) (\Pi P_\perp V)^\top. \quad (10)$$

A $P \times P$ target image T has P^2 numbers ($\approx 10^5$); Z offers $\approx D(D - N) \approx 3 \times 10^{11}$ free parameters. The counting is overwhelmingly favorable — the constraint is not the image, it is only that a rank- N sliver ($P_\perp$) is reserved for the music. Concretely, we want to choose U, V such that

$$(\Pi U)(\Pi P_\perp V)^\top = T_{\text{amp}} - M_0 \equiv \Delta, \quad (11)$$

where T_{amp} is the target after centering and amplitude scaling (display is affine-invariant, so this loses nothing).

Step 1: factor the image (SVD). The truncated singular value decomposition

$$\Delta \approx W_q \Sigma_q Y_q^\top = (W_q \Sigma_q^{1/2})(Y_q \Sigma_q^{1/2})^\top \equiv \tilde{U} \tilde{V}^\top, \quad \tilde{U}, \tilde{V} \in \mathbb{R}^{P \times q}, \quad (12)$$

is the best rank- q approximation of the residual image (Eckart–Young). Note the SVD’s role: it is a *compression* of the target into q outer products — nothing about null spaces. Smooth, cloud-like images need small q ; images with sharp lines need larger q . (Measured: 0.6% error at $q = 160$ for a cloud target; up to 9% for crack-network targets at $q = 220$.)

Step 2: lift pixels to samples. We need sample-level columns $U \in \mathbb{R}^{D \times q}$ whose pooled version is $\tilde{U}$, i.e. $\Pi U = \tilde{U}$. Since Π averages k samples into one pixel, any right inverse works; we use smooth linear interpolation between block centers plus one exact piecewise-constant correction, so $\Pi(\text{lift}(\tilde{U})) = \tilde{U}$ holds *exactly* while the columns stay smooth. Same for $\tilde{V} \rightarrow V$.

Step 3: the projection correction (fixed-point iteration). One wrinkle remains: the picture uses $\Pi P_{\perp} V$, not ΠV , and

$$\Pi P_{\perp} V = \Pi V - \Pi X G^{-1} X^{\top} V \quad (13)$$

differs from $\Pi V = \tilde{V}$ by the pooled shadow of a *rank- N* term — small, because generic image columns are nearly orthogonal to the N songs, but not zero. We remove it by iterating

$$V \leftarrow V + \text{lift}(\tilde{V} - \Pi P_{\perp} V), \quad (14)$$

which is a classical fixed-point (Richardson) iteration: writing the error $e = \tilde{V} - \Pi P_{\perp} V$, one step maps $e \leftarrow \Pi X G^{-1} X^{\top} \text{lift}(e)$, an operator of rank $\leq N$ and small norm, so the error contracts geometrically; 2–3 iterations reach sub-percent accuracy. (This is *not* power iteration — power iteration amplifies a dominant eigenvector; here we drive a small residual to zero, the same principle as iterative refinement in linear solvers.)

Result. With U and the corrected $P_{\perp} V$ in (7), A still plays the playlist exactly (anything through $P_{\perp}$ is inaudible), and its block-mean picture equals the chosen image to within the SVD truncation. The aesthetic search then moves entirely into ordinary 2D image space: generate candidate images procedurally, score them with perceptual metrics whose weights are fitted to the owner by pairwise comparison (Bradley–Terry), and embed the winner. The mathematics guarantees the two roles never interfere: *the music constrains A on an N -dimensional subspace; the image lives on the complement.*

Summary in one line

$$A = \underbrace{X_{\text{next}} G^{-1} X^{\top}}_{\text{plays the playlist}} + \underbrace{U(P_{\perp} V)^{\top}}_{\text{draws the image}}, \quad Ax_n = x_{n+1} \text{ exactly}, \quad \Pi A \Pi^{\top} \approx T.$$